

NEWINGTON COLLEGE



2014 HSC Assessment 1 Year 12 Mathematics Extension 1

General Instructions:

- Date of task – Monday 24th November (Wk 8B)
- Working time – 45 mins
- Weighting - 15%
- Board-approved calculators may be used.
- Attempt all questions, start each question in a new booklet.
- Show all relevant mathematical reasoning and/or calculations.

Outcome	Marks
Section 1 – Multiple choice	/4
Section 2 – Differentiation and Integration	/11
Section 3 – Area, Volume and Curve Sketching	/15
Total	/30

Outcomes to be assessed:

- PE5** Determines derivatives which require the application of more than one rule of differentiation.
- HE5** Applies the chain rule to problems appropriate techniques from the study of series to solve problems

Section 1: Multiple Choice (4 Marks)

1) Which expression is equal to $\int \frac{4x}{3x^2+1} dx$?

(A)

(B)

(C)

(D)

$$\frac{2}{3}\ln(3x^2 + 1) + c \quad \frac{3}{2}\ln(3x^2 + 1) + c \quad 4\ln(3x^2 + 1) + c \quad \ln\left(\frac{4}{6x}\right) + c$$

2) If $f(x) = 2^x$ then $f'(x)$ is equal to:

(A)

(B)

(C)

(D)

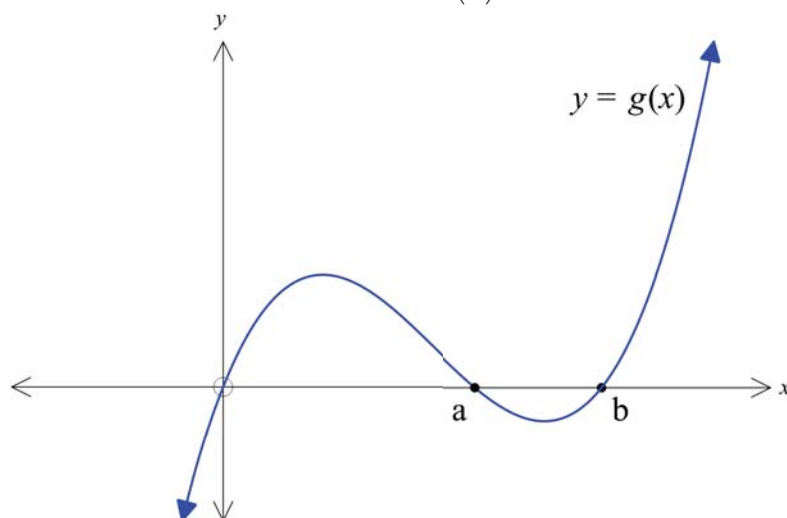
$\ln 4^x$

$\ln 2 \times 2^x$

$x \times 2^x$

$\ln 2^x \times 2^x$

3) The graph of the function $y = g(x)$ is shown below.



Which expression **DOES NOT** correctly describe the area bounded by the $y = g(x)$ and the x axis, between $x = 0$ and $x = b$?

(A)

(B)

$$\int_0^a g(x) dx - \int_b^a g(x) dx$$

$$\left| \int_0^a g(x) dx \right| + \int_b^a g(x) dx$$

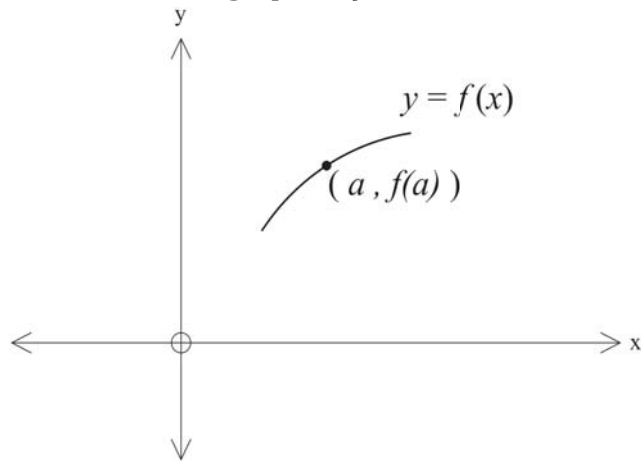
(C)

(D)

$$\left| \int_0^a g(x) dx + \int_a^b g(x) dx \right|$$

$$\left| \int_a^0 g(x) dx \right| + \left| \int_a^b g(x) dx \right|$$

4) A section of the graph of $f(x)$ is shown below.



(A)

$$f'(x) < 0$$

$$f''(x) < 0$$

(B)

$$f'(x) > 0$$

$$f''(x) < 0$$

(C)

$$f'(x) < 0$$

$$f''(x) > 0$$

(D)

$$f'(x) > 0$$

$$f''(x) > 0$$

Section 2: Differentiation and Integration (11 Marks)

1) (a) Find $\int \frac{4x^2 - 3x}{x} dx$ 2

(b) Find $\frac{d}{dx} \left(\frac{e^x}{x^2} \right)$ 2

(c) Find $\int z \cdot \sqrt[3]{z^2 + 1} dz$ by using the substitution $u = z^2 + 1$ 3

(d) (i) Show that $\frac{5x - 4}{x - 2} = 5 + \frac{6}{x - 2}$ 1

(ii) Hence, or otherwise, evaluate 3

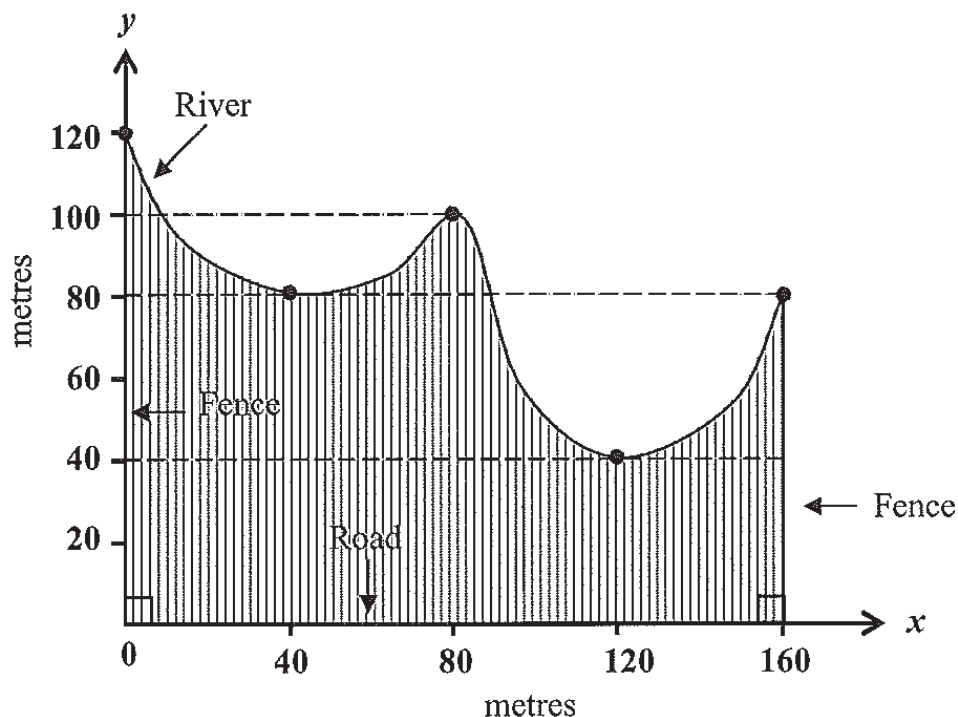
$$\int_3^4 \frac{10x - 8}{x - 2} dx$$

Section 3: Area, Volume and Curve Sketching (15 Marks)

1) Find the area bounded by the curves $f(x) = x^2$ and $g(x) = 3x - 2$ 3

2) Find the volume of the solid generated when $y = e^x - \frac{1}{e^x}$ is rotated about the x -axis between $x = 0$ and $x = 0.5$. Leave your answer in simplest exact form. 3

- 3) A paddock is bounded by 3 straight sides and a river, as shown by the shaded area in the scale diagram below.



- (i) By reading distances from the diagram and using the **trapezoidal rule** with 5 function values, find the area of the paddock. 2
- (ii) Does the trapezoidal rule give an under-estimate or an over-estimate for the actual area? Justify your answer 1
- 4) For the curve $y = \frac{x}{x^2 + 1}$, 6
- Show that there are turning points at $\left(1, \frac{1}{2}\right)$ and $\left(-1, -\frac{1}{2}\right)$ and determine their nature.
 - Show that points of inflexion occur when $x = 0$ and $x = \pm\sqrt{3}$.
 - State the equation of the horizontal asymptote.
 - Sketch the curve, labeling all important features.

END OF EXAMINATION

Student Name:..... Number: Teacher:.....

Year 12 Extension 1
Section 1 – Multiple Choice Answer Sheet

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|---|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

HSC ASSESSMENT 1 - YEAR 12 EXT 1

Section 1

1. $\frac{d}{dx} \frac{4x}{3x^2+1}$

$$= 4 \cdot \frac{d}{dx} \frac{x}{3x^2+1}$$

$$= \frac{2}{3} \frac{d}{dx} \frac{6x}{3x^2+1}$$

$$= \frac{2}{3} \ln(3x^2+1)$$

A

2. $y = 2^x$

$$\ln y = x \ln 2$$

$$x = \frac{\ln y}{\ln 2}$$

$$\frac{dx}{dy} = \frac{1}{y \ln 2}$$

$$\frac{dy}{dx} = y \ln 2$$

$$= \ln 2 \cdot 2^x$$

B

3. C

4. B

Section 2

1. a) $\int \frac{4x^2 - 3x}{x} dx$

$$= \int 4x - 3 dx$$

$$= \frac{4x^2}{2} - 3x + c$$

$$= 2x^2 - 3x + c$$

$$b) \quad \frac{d}{dx} \left(\frac{e^x}{x^2} \right) \quad u = e^x \quad v = x^2$$

$$u' = e^x \quad v' = 2x$$

$$= \frac{x^2 \cdot e^x - e^x \cdot 2x}{x^4}$$

$$= \frac{x e^x - 2 e^x}{x^3}$$

$$= \frac{e^x (x - 2)}{x^3} //$$

$$c) \quad \int \frac{z}{z^2 + 1}^{\frac{1}{3}} dz \quad u = z^2 + 1$$

$$\frac{du}{dz} = 2z$$

$$dz = \frac{du}{2z}$$

$$= \int \frac{z}{z^2 + 1}^{\frac{1}{3}} \cdot \frac{du}{2z}$$

$$= \frac{1}{2} \int u^{\frac{1}{3}} du$$

$$= \frac{1}{2} \left[\frac{3}{4} u^{\frac{4}{3}} \right] + C$$

$$= \frac{3}{8} (z^2 + 1)^{\frac{4}{3}} //$$

$$d) i) \quad \frac{5x - 4}{x - 2} = \frac{5x + 10 + 6}{x - 2} \quad ii) \quad = \int_3^4 \frac{5 + 6}{x - 2} dx$$

$$= \frac{5x - 10}{x - 2} + \frac{6}{x - 2}$$

$$= \frac{5(\cancel{x} - 2)}{\cancel{x} - 2} + \frac{6}{x - 2}$$

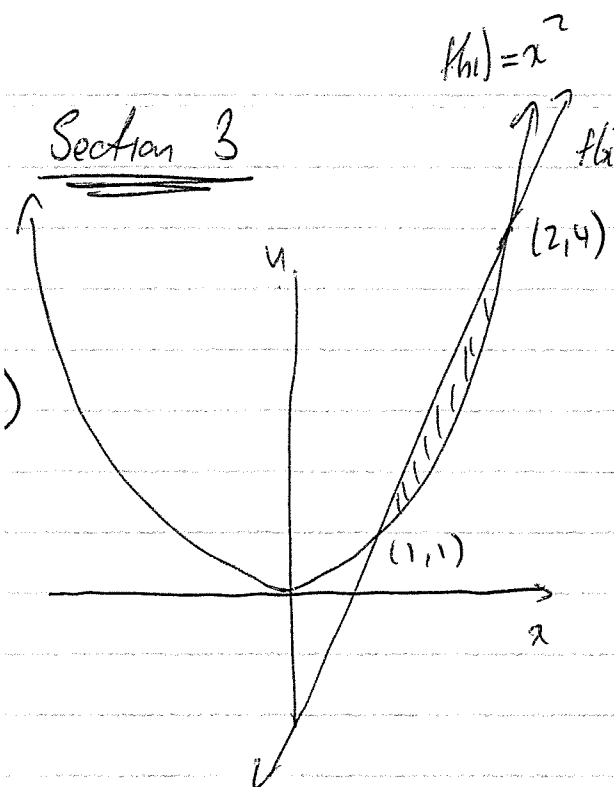
$$= 5 + \frac{6}{x - 2} //$$

$$= \left[5x + 6 \ln(x - 2) \right]_3^4$$

$$= (20 + 6 \ln 2) - (15 + 6 \ln 1)$$

$$= 5 + 6 \ln 2 //$$

Section 3



$f(x) = 3x - 2$ Intersection:

$$x^2 = 3x - 2$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 2 \quad x = 1$$

$$y = 4 \quad x = 1$$

$$\text{Area} = \int_1^2 (3x-2) - (x^2) dx$$

$$= \left[\frac{3x^2}{2} - 2x - \frac{x^3}{3} \right]_1^2$$

$$= \left(\frac{12}{2} - 4 - \frac{8}{3} \right) - \left(\frac{3}{2} - 2 - \frac{1}{3} \right)$$

$$= \frac{1}{6} u^2$$

$$2) \quad V = \pi \int f(x)^2 dx$$

$$\left(\frac{e^x - 1}{e^x} \right)^2 =$$

$$V = \pi \int_0^{\frac{1}{2}} \left(e^x - \frac{1}{e^x} \right)^2 dx$$

$$= \pi \int_0^{\frac{1}{2}} (e^{2x} - 2 + e^{-2x}) dx$$

$$= \pi \left[\frac{e^{2x}}{2} - 2x - \frac{e^{-2x}}{2} \right]_0^{\frac{1}{2}}$$

$$= \pi \left[\left(\frac{e}{2} - 1 - \frac{1}{2e} \right) - \left(\frac{1}{2} - 1 - \frac{1}{2} \right) \right]$$

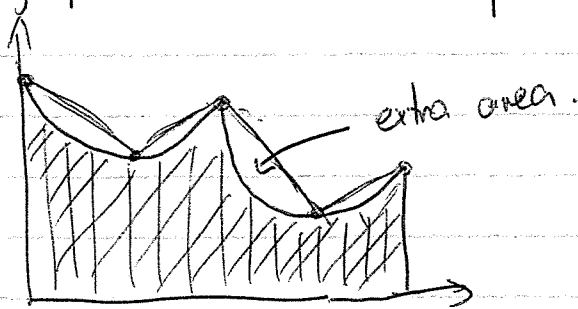
$$= \pi \left(\frac{e}{2} - 1 - \frac{1}{2e} \right) u^3$$

3) i) $h = 40$

$$A \approx \frac{40}{2} (1 \times 120 + 2 \times 80 + 2 \times 100 + 2 \times 40 + 1 \times 160)$$

$$\approx 14400 \text{ m}^2$$

ii) This would be an overestimation as graph is concave up.



4.) i) $y = \frac{x}{x^2+1}$ $u=x$ $v=x^2+1$
 $u'=1$ $v'=2x$

$$y' = \frac{(x^2+1) \cdot 1 - x \cdot 2x}{(x^2+1)^2}$$

$$= \frac{x^2+1-2x^2}{(x^2+1)^2}$$

$$= \frac{1-x^2}{(x^2+1)^2}$$

Turning pts at $y' = 0$

$$0 = \frac{1-x^2}{(x^2+1)^2}$$

$$0 = (1-x)(1+x)$$

$$x=1$$

$$x=-1$$

$$y = \frac{1}{2}$$

$$y = -\frac{1}{2}$$

x	-2	-1	0	1	2
y'	$-\frac{3}{25}$	0	1	0	$-\frac{3}{25}$

\ - / - \

← Bu table or y''

ii) Inflection at $y''=0$

$$y' = \frac{1-x^2}{(x^2+1)^2}$$

$$u = 1-x^2 \\ u' = -2x$$

$$v = (x^2+1)^2 \\ v' = 2(x^2+1)(2x)$$

$$y'' = \frac{(x^2+1)(-2x) - (1-x^2)(2(x^2+1)(2x))}{(x^2+1)^2} = 0$$

$$2x(x^2+1)[- (x^2+1) - 2(1-x^2)] = 0$$

$$2x(x^2+1)(x^2-3) = 0$$

$$\begin{matrix} / & & / \\ x=0 & & x=\pm\sqrt{3} \end{matrix} \quad (y=\pm\frac{1}{8})$$

$$\begin{aligned} \text{iii) } y &= \frac{x}{x^2+1} & \lim_{x \rightarrow \infty} \frac{x}{x^2+1} &= \lim_{x \rightarrow \infty} \frac{x/x^2}{x^2/x^2 + 1/x^2} \\ & & &= \lim_{x \rightarrow \infty} \frac{1/x}{1 + 1/x^2} \\ & & &= 0 \end{aligned}$$

Horizontal asymptote is $y=0$

iv)

